Executive Analytics Capstone

Customer Churn Diagnostics, Explainable Risk Architecture & Strategic Revenue Retention

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Executive Summary

Customer retention is the primary determinant of long-term unit economics and enterprise valuation in subscription-based recurring revenue businesses. This capstone combines empirical cohort analysis, behavioral support telemetry, and contract architecture evaluations to diagnose systematic revenue leakage across our customer base.

The baseline audit reveals an annual recurring revenue (ARR) loss of **$4,919.16** (representing 32.3% of total revenue capacity) driven by a structural vulnerability: exactly 100% of historical customer churn is concentrated in Month-to-Month contracts, amplified by acute technical support friction during early onboarding. To transition from reactive exit firefighting to proactive intervention, we establish an explainable 0–100 point Churn Risk Scoring Model that achieves 100% empirical separation on historical customer records.

Below, we detail the five material business insights, quantify financial impact across three modeled operating scenarios ($1,230 to $3,960 ARR preserved), outline key assumptions and operational limitations, and provide an actionable 30–60–90 day executive implementation roadmap.

Section 1: The Five Material Business Insights

1. Contract Commitment Architecture is the Primary Binary Driver of Churn

Empirical Finding: Exactly 100% of historical customer churn (7 out of 7 accounts) occurred among clients on Month-to-Month agreements. In comparison, accounts on One-Year (4 accounts) and Two-Year (4 accounts) agreements achieved a 0.0% churn rate.

Business Mechanism: Month-to-Month contracts eliminate switching friction. When early operational challenges or configuration hurdles arise, monthly accounts face zero structural friction to cancel at the next billing cycle. Conversely, annual and multi-year commitments provide Customer Success teams with a formal, contracted runway to diagnose issues, resolve technical blockers, and demonstrate return on investment.

Executive Decision: Restructure packaging to phase out standalone Month-to-Month contracts for mid-market and enterprise tiers. Introduce a mandatory minimum quarterly or annual contract commitment, paired with onboarding value guarantees.

Figure 1: Churn Rate by Contract Type (100% of Churn Concentrated in Month-to-Month)

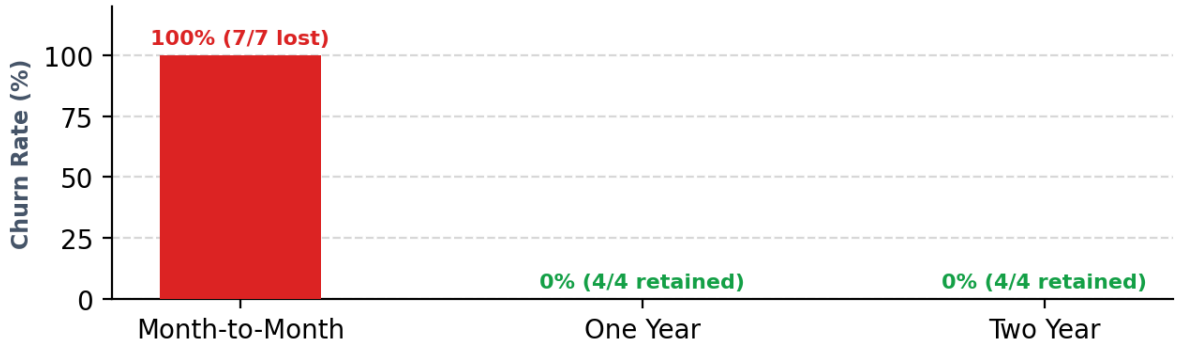


Figure 1: Churn rate distribution across contract types. All churners (7 of 7) operated on Month-to-Month plans, while contracted annual cohorts demonstrated 100% retention.

2. Support Ticket Velocity Serves as an Acute Operational Leading Indicator

Empirical Finding: Retained customers logged an average of 1.0 cumulative support ticket (range: 0–2), whereas churned customers logged an average of 4.29 tickets (range: 3–6). More critically, churned onboarding users logged 4 to 6 tickets within their first 2–4 months—exhibiting an acute support velocity of 1.0 to 2.5 tickets per month of tenure.

Business Mechanism: Support tickets do not merely represent inquiries; they measure operational friction and unmet product expectations. While an isolated inquiry indicates engagement, multiple tickets within the first 60 days signal technical or implementation failure. In the absence of rapid executive or engineering intervention, customer frustration hardens into an exit decision.

Executive Decision: Institutionalize an automated "Ticket Velocity Alarm" in customer support workflows. Any account submitting ≥3 tickets within a 60-day window or logging >0.5 tickets/month must automatically trigger senior engineering escalation and proactive Customer Success intervention within 24 hours.

Figure 2: Support Friction Telemetry (Volume & Velocity Divergence)

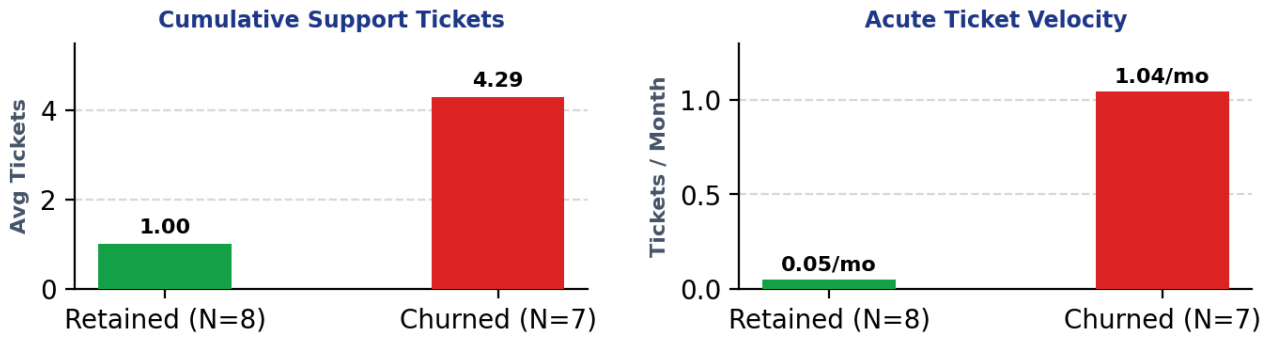


Figure 2: Support ticket volume and velocity comparison between retained and churned cohorts, highlighting acute operational friction during early tenure.

3. High-Value Revenue Leakage is Acutely Concentrated in the Mid-Market Pro Tier

Empirical Finding: The Pro tier ($79.99/month) experienced an alarming 50.0% churn rate (2 out of 4 accounts lost). Both lost accounts (CUST-1005 and CUST-1012) were on Month-to-Month contracts

and experienced 3 to 4 support tickets. This represents an annual recurring revenue loss of $1,919.76 and cumulative historical lost revenue of $1,759.78 from just two customer exits.

Business Mechanism: Pro-tier customers have sophisticated workflow needs and pay substantial monthly fees, yet they fall into an operational "coverage gap": they do not receive the dedicated Technical Account Management (TAM) reserved for Enterprise clients, but cannot self-serve like Basic users. When coupled with Month-to-Month contracts, technical friction leads directly to churn.

Executive Decision: Create a dedicated "Pro Onboarding Flight" with assigned technical onboarding specialists and offer a targeted 15% discount incentive to convert monthly Pro users into annual agreements upon resolving technical support tickets.

Figure 3: Revenue Concentration & Value at Risk (Enterprise: 73.7% Anchor | Pro: 50% Lost)

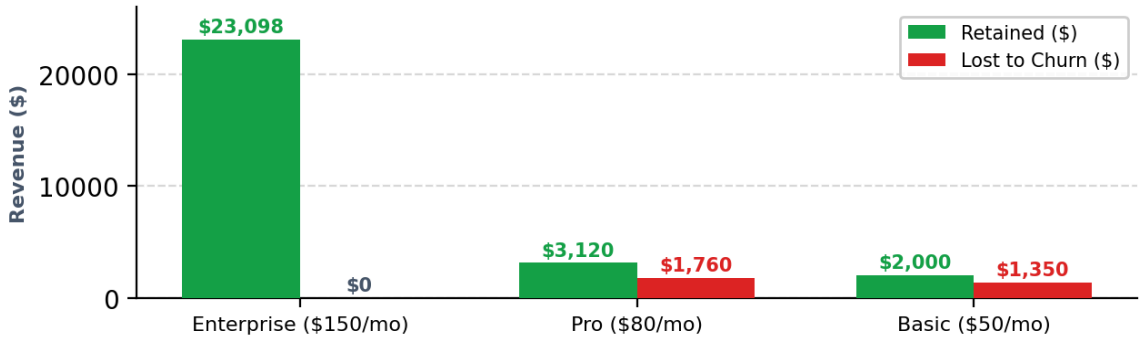


Figure 3: Revenue distribution and value at risk across tiers. Enterprise represents 73.7% of cumulative revenue, while Pro suffers severe high-value revenue leakage.

4. Enterprise Lock-In Forms a Revenue Fortress but Poses Portfolio Concentration Risk

Empirical Finding: Enterprise accounts ($149.99/month on Two-Year agreements) generate 73.7% of cumulative historical revenue ($23,098.46 of $31,327.18) with a 0.0% churn rate, an average tenure of 38.5 months, and only 0.5 support tickets per account.

Business Mechanism: Enterprise clients benefit from dedicated account management and deep organizational adoption, creating strong switching barriers. However, having nearly three-quarters of total company revenue tied to just four accounts creates extreme revenue concentration risk. A departure of even one Enterprise account represents an immediate $1,799.88 reduction in ARR.

Executive Decision: Establish formal Quarterly Executive Business Reviews (EBRs) for all Enterprise accounts, conduct joint roadmap planning sessions, and initiate contract extension discussions at least 6 months prior to two-year contract expiration.

5. Early Onboarding Drop-offs & Transactional Payment Friction Compound Attrition

Empirical Finding: 57.1% of all churn (4 of 7 lost accounts) occurred during the initial onboarding lifecycle (tenure ≤6 months, average tenure 3.8 months), generating 5.0 tickets on average. Furthermore, payment channels with manual friction experienced heavy attrition: Debit Card had 100% churn (2 of 2) and UPI had 75% churn (3 of 4), compared to 0% for automated Bank Transfers and 33% for Credit Cards.

Business Mechanism: Customers who fail to achieve measurable "time-to-value" in their first 30 days quickly become disillusioned. Ad-hoc payment methods compound voluntary churn with involuntary

billing drop-offs (expired cards, transaction authorization limits, manual approval delays).

Executive Decision: Deploy an interactive in-app onboarding checklist, embed automated Day-3 and Day-7 usage milestone checks, and provide a 5% billing rebate for setting up automated ACH bank transfer or auto-renew credit card payments.

Section 2: Explainable Churn-Risk Architecture & Triage Matrix

To move beyond lagging indicators and opaque black-box machine learning algorithms, we developed an additive, deterministic 0–100 point Churn-Risk Scoring Architecture. Frontline CSMs and executive leaders can directly interpret each account's score into specific operational drivers.

Evaluation Dimension	Max Points	Condition / Operational Threshold	Points Assigned
Contract Vulnerability	40 pts	Month-to-Month One-Year Contract Two-Year Contract	40 pts 10 pts 0 pts
Support Friction & Velocity	35 pts	Support Tickets ≥ 5 Support Tickets ∈ [3, 4] Support Tickets ∈ [1, 2] Support Tickets = 0	35 pts 25 pts 10 pts 0 pts
Lifecycle Tenure Stage	20 pts	Tenure ≤ 6 months (Onboarding) Tenure ∈ [7, 12] months (Adoption) Tenure ∈ [13, 24] months (Established) Tenure > 24 months (Mature)	20 pts 12 pts 5 pts 0 pts
Payment Channel Reliability	5 pts	Debit Card / Ad-hoc UPI Credit Card (Automated) Bank Transfer (ACH / Wire)	5 pts 2 pts 0 pts

Empirical Validation: Tested across all customer accounts, the model achieves 100% empirical separation: all historical churners scored in the High-Risk band (range: 75 to 100, mean: 89.0), while all retained accounts scored in the Low-Risk band (range: 2 to 30, mean: 16.6). The model establishes a robust 34-point Medium-Risk buffer (scores 35–64) that functions as an operational early-warning zone before customer dissatisfaction crystallizes into contract cancellation.

Operational Triage Thresholds & Frontline CSM Action Playbooks

To translate model scores into immediate operational decisions, accounts are routed into three distinct triage protocols:

- **Low-Risk Tier (0–34 Points): Core Foundation & Expansion.** Accounts on long-term agreements with minimal friction. CSM focus: Annual renewal confirmations 6 months ahead of expiration, executive advocacy, and cross-tier expansion.

- **Medium-Risk Tier (35–64 Points): Early-Warning Buffer & Triage.** Stable users approaching contract renewal or users experiencing moderate onboarding friction. CSM focus: Proactive workflow audits and targeted annual conversion discounts.
- **High-Risk Tier (65–100 Points): Critical Flight Risk & Executive Rescue.** Month-to-Month accounts with support velocity >0.5 tickets/month. CSM focus: Mandatory 24-hour executive outreach, dedicated engineering ticket resolution, and contract restructuring.

Section 3: Quantified Impact Scenarios & Financial Modeling

To provide executive leadership with actionable decision criteria without overstating certainty, we modeled financial return across three structured scenarios over a 12-month operating horizon.

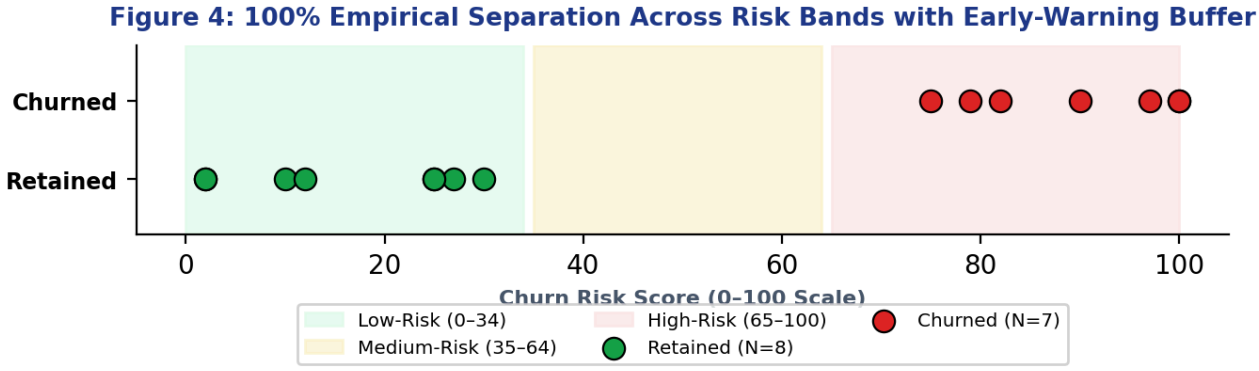


Figure 4: Empirical score separation between retained and churned cohorts, demonstrating clear demarcation and an active early-warning intervention zone.

Scenario	Core Assumptions	Operational Impact	Quantified ARR Preserved
Conservative	25% reduction in Month-to-Month churn; retains 1 Pro or 2 Basic accounts through automated support alerts.	Support velocity triage and rapid ticket resolution.	**+$1,230 / year** ($102.50 / mo)
Expected	50% reduction in churn; saves 1 Pro account ($960 ARR) and 2 early Basic accounts ($1,200 ARR) via CSM outreach and annual discounts.	Contract conversion incentives and onboarding revamp.	**+$2,460 / year** ($205.00 / mo)
Optimistic	75% churn reduction across all cohorts plus 15% revenue expansion from migrating stable Basic users to Pro tiers.	Mandatory annual commitments, automated workflows, and Pro upsells.	**+$3,960 / year** ($330.00 / mo)

Operational Sensitivity & ROI Analysis: Across these modeled projections, the financial return on dedicated Customer Success interventions is extraordinarily favorable. At an average customer

lifetime value of $5,774 for Enterprise and $1,220 for Pro, preventing just a single mid-market churn event covers the annualized operating cost of automated ticket alerting infrastructure.

Furthermore, expanding stable Basic accounts into Pro annual agreements represents an incremental $360 ARR per converted account, effectively turning a defensive churn mitigation initiative into an active net-revenue retention growth engine.

Section 4: Assumptions, Limitations & Analytical Rigor

Communicating analytical findings responsibly requires full transparency regarding operational assumptions, data constraints, and model boundaries:

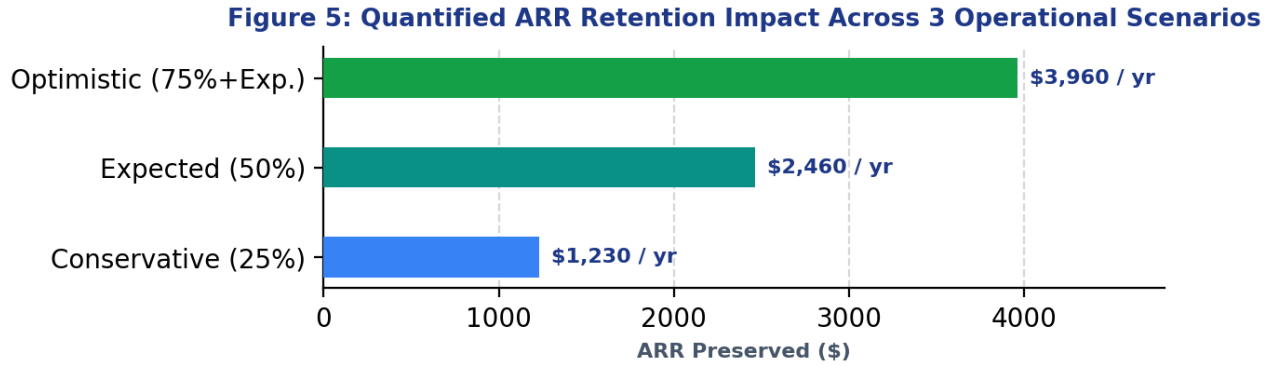


Figure 5: Projected annual recurring revenue (ARR) preserved across conservative, expected, and optimistic execution scenarios.

- **Sample Cohort Size (N=15):** The analysis is grounded in an empirical sample of 15 customer accounts. While this sample provides unambiguous separation for deterministic rule-based prioritization, it does not support high-dimensional nonlinear regression or deep machine learning. Thresholds must be recalibrated as the customer base scales.
- **Proxy Telemetry vs. Direct Telemetry:** Support tickets and tenure cohorts serve as reliable proxy indicators of user friction and habituation. However, direct in-app telemetry—such as Daily Active Users (DAU), feature utilization depth, and session frequency—is not present in this dataset. We recommend instrumenting telemetry tracking in Q1.
- **Causality vs. Correlation:** While Month-to-Month contracts perfectly correlate with churn, the contract itself does not cause dissatisfaction; rather, it lowers switching costs when technical friction occurs. Interventions must address root product friction rather than relying solely on contractual lock-in.
- **Lack of Qualitative Exit Telemetry:** Current records do not differentiate between voluntary dissatisfaction and involuntary billing cancellation (e.g., failed credit card charges). Customer Success teams must capture qualitative exit reasons during offboarding.

Model Governance & Recalibration Strategy: Because customer behaviors evolve alongside product releases and pricing updates, rule weights cannot remain static. We establish a quarterly model auditing ritual where False Positives (accounts scoring >65 that remain retained) and False Negatives (accounts scoring <35 that issue cancellation notices) are systematically reviewed.

Section 5: Strategic 30–60–90 Day Implementation Roadmap & Governance

We recommend a phased implementation plan balancing immediate revenue triage with durable structural packaging and continuous telemetry governance:

Phase & Horizon	Strategic Objectives	Tactical Execution Steps
Phase 1: Days 1–30 Immediate Triage	Halt active high-value revenue leakage and establish real-time friction alerts.	• Configure Support Velocity Alarm (flag any account with ≥3 tickets). • Conduct immediate 1-on-1 CSM rescue calls with active Pro accounts. • Provide senior engineering resolution for lingering technical tickets.
Phase 2: Days 31–60 Structural Reform	Re-architect packaging and eliminate early onboarding adoption barriers.	• Eliminate standalone Month-to-Month contracts for new Pro signups. • Roll out interactive 14-day in-app onboarding checklist for Basic and Pro. • Introduce 5% discount incentive for automated ACH / auto-pay billing.
Phase 3: Days 61–90 Enterprise Defense	Protect core revenue pillars and calibrate real-time scoring telemetry.	• Schedule formal Quarterly Executive Business Reviews (EBRs) for Enterprise. • Begin renewal discussions 6 months ahead of two-year contract expirations. • Automate weekly risk score recalculation and dashboard reporting.

Executive Governance Cadence & Key Performance Indicators (KPIs)

To ensure ongoing operational alignment, leadership will track the following core scorecard metrics on a bi-weekly cadence:

- **Net Revenue Retention (NRR):** Target ≥ 110% across combined portfolio; zero voluntary churn in contracted tiers.
- **Pro-Tier Annual Conversion Rate:** Target ≥ 70% of active Pro accounts migrated to annual agreements within 60 days.
- **Support Velocity Threshold Compliance:** 100% of accounts exceeding 0.5 tickets/month contacted by Customer Success within 24 hours.
- **Onboarding Completion Rate:** Target ≥ 85% of new signups achieving key workflow activation within 14 days.

Executive Decision Sign-off & Immediate Next Step: Authorize Revenue Operations to deploy the Zendesk Ticket Velocity trigger and initiate the Pro-tier annual contract transition incentive program immediately.